Campus Network Design and Implementation Report

Name	Muhammad Hakim Bin Zaaba
Started at	17 Jan 2026 (Sunday)
Finished at	21 Jan 2026 (Wednesday)

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Phase 1 : Network Design

1. Requirements Gathering

The objective of this project is to design and implement a realistic campus network using a hierarchical model. The network must support multiple user groups, redundancy, routing, security policies, and voice services. The following requirements were identified:

- Segmentation of users using VLANs
- Redundant default gateways using HSRP
- Dynamic routing using OSPF
- Centralized network services (DHCP, VoIP)
- Secure guest access with access control
- High availability through multiple paths and devices

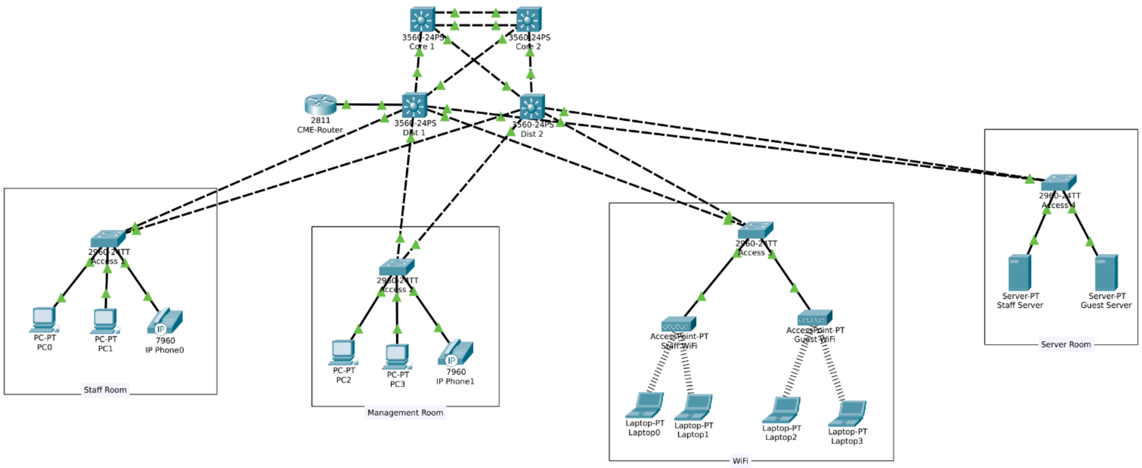
VLAN requirements:

- VLAN 10 : Staff
- VLAN 20 : Guest
- VLAN 30 : VoIP
- VLAN 90 : Management

Redundancy requirements:

- Dual core switches
- Dual distribution switches
- Access switches dual-homed to distribution layer

Topology:



2. Network Diagram

The network follows a three-tier hierarchical model:

- Core Layer: Provides fast, policy-free Layer 3 transport
- Distribution Layer: Performs inter-VLAN routing, gateway redundancy, and policy enforcement
- Access Layer: Connects end devices and applies VLAN membership

The VoIP Call Manager (CME Router) and centralized services are placed at the distribution layer to reflect real-world enterprise practices.

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3. Hardware / Software Specifications

Hardware

- Core Switches: Cisco Multilayer Switches (2 units)
- Distribution Switches: Cisco Multilayer Switches (2 units)
- Access Switches: Cisco Layer 2 Switches (4 units)
- Router: Cisco Router running CME
- End Devices: PCs, IP Phones
- Server: Centralized DHCP / DNS Server

Software

- Cisco Packet Tracer 9.0
- Cisco IOS (Multilayer Switch & Router)

Phase 2 : Network Implementation

4. Device Configuration

Each device was configured according to its role in the network hierarchy.

- Core switches: IP routing and OSPF only
- Distribution switches: SVIs, HSRP, ACLs, OSPF, DHCP relay
- Access switches: VLANs, trunk links, access ports
- CME Router: Call Manager Express configuration

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5. Interface Configuration

- Trunk links configured between Access and Distribution switches
- Routed links configured between Distribution and Core switches
- Core-to-Core routed links configured for redundancy
- Voice VLAN enabled on access ports connected to IP phones

OSPF point-to-point network type was used on routed links to simplify adjacency and improve convergence.

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6. Global and Management Configuration

- IP routing enabled on multilayer switches
- OSPF configured with passive interfaces on VLAN SVIs
- HSRP configured on all VLAN gateways
- Management VLAN (VLAN 90) used for device management access

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7. DHCP Configuration

A centralized DHCP server was deployed in VLAN 10. Distribution switches use DHCP relay (`ip helper-address`) to forward DHCP requests from other VLANs.

DHCP scopes:

- VLAN 10: Staff
- VLAN 20: Guest
- VLAN 30: VoIP (with Option 150 for CME)
- VLAN 90: Management

This design avoids DHCP duplication and reflects realistic enterprise architecture.

Phase 3 : Network Testing and Verification

8. Connectivity Testing

Tests performed:

- Inter-VLAN communication
- Internet path simulation via core layer
- Redundancy testing by shutting down links and devices

Results:

- OSPF successfully rerouted traffic during link failures
- HSRP maintained gateway availability
- End-user connectivity experienced minimal disruption

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9. VLAN Verification

Verification commands used:

- show vlan brief
- show interfaces trunk
- show ip interface brief

Results confirmed:

- Correct VLAN assignment
- Proper trunk operation
- Active SVIs on distribution switches

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10. VoIP Testing

- IP phones successfully obtained IP addresses via DHCP
- Option 150 correctly pointed to CME router
- Phones registered with Call Manager
- Calls between IP phones were successful

VoIP failure testing confirmed that network connectivity remains intact even if CME service is unavailable.

Phase 4 : Documentation

11. Network Diagram Documentation

The final network diagram clearly illustrates:

- Device roles and hierarchy
- VLAN segmentation
- Redundant paths
- Service placement

This diagram can be used for operational reference and troubleshooting.

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12. Configuration Documentation

All configurations were documented per device, including:

- VLAN and interface configuration
- OSPF routing configuration
- HSRP settings
- ACL rules
- DHCP relay configuration

This documentation enables quick recovery and audit of the network.

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13. Troubleshooting Log

Key issues encountered and resolved:

1. DHCP failure due to ACL blocking UDP ports 67/68
    - Resolved by explicitly permitting DHCP traffic
2. IP phones stuck at "No CM List"
    - Resolved by configuring DHCP Option 150 and CME router
3. Guest VLAN bypassing ACL during HSRP failover
    - Resolved by mirroring ACLs on both distribution switches
4. Blackhole scenario during uplink failure
    - Identified need for HSRP tracking (design improvement)

Conclusion

This project successfully demonstrates the design, implementation, and verification of a realistic campus network. The network meets all functional and redundancy requirements while following best-practice enterprise design principles. The troubleshooting process further strengthened understanding of protocol roles, failure domains, and operational behavior in real networks.